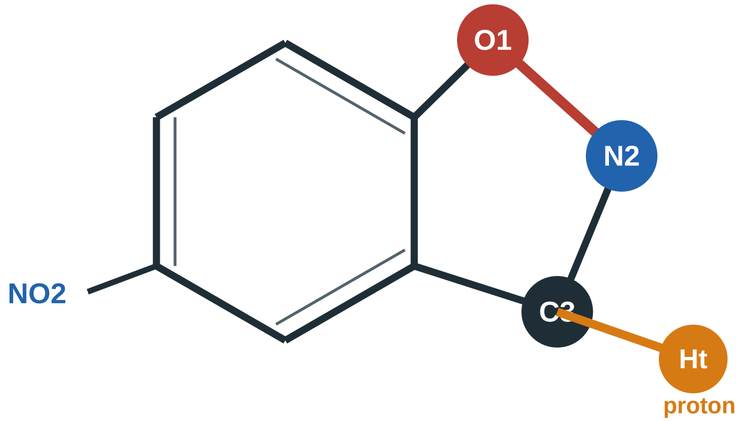
# Kemp elimination: proton transfer opens the ring

Asp162 accepts a substrate proton as the N-O bond breaks and the product forms.

## REACTANT STATE

**5-nitrobenzisoxazole**




**Asp162**
carboxylate

OD1 **accepts Ht**

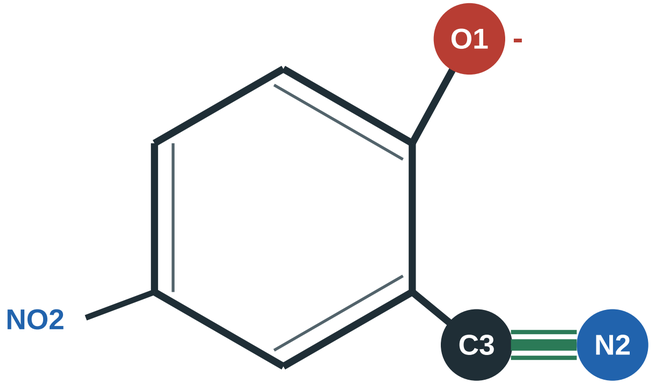
**ONE STEP**

**concerted**
bond changes

**C3-Ht -> OD1-Ht**
**N2-O1 breaks**
**C3-N2 becomes nitrile**

## PRODUCT STATE

**2-cyano-4-nitrophenoxide**




**Asp162**
now protonated

OD1 **- Ht**

**Same atoms; different bonds and proton placement.**

Schematic only, not to scale. The paper describes a concerted reaction path; no separate stable intermediate is shown.